

MASTER VENDOR DOSSIER

Next-Generation Enterprise University Management System

Prepared Exclusively For:

K.S.R.M. College of Engineering (Autonomous)

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Authored by:

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Lead Enterprise Architect

1. System Architecture & Capabilities

K.S.R.M. College of Engineering requires a digital infrastructure that matches its autonomous prestige. This proprietary University Management System (UMS) replaces fragmented legacy systems with a highly secure, Next.js serverless architecture engineered specifically for the Kadapa campus.

1.1 Enterprise Modules

- **Controller of Exams (EMS):** Automated SGPA calculations, instant hall ticket generation, and tamper-proof grading ledgers built directly on Prisma transactional logic.
- **Automated Finance Ledger:** Direct API integration simulating student payment gateways (Razorpay) seamlessly converting to administrative XML outputs for immediate Tally ERP synchronization.
- **Hardware-Agnostic Attendance:** Real-time attendance transaction engine designed to prevent data desynchronization during network instability, complete with automated condonation warnings.
- **Disciplinary & Governance Engine:** Immutable, time-stamped audit trails for faculty-to-student disciplinary loops. Provides instant, 1-click reporting critical for passing NAAC Criteria 5 inspections.

1.2 Zero-Trust Security & Scale Defense

The "Results Day" Guarantee: Standard college ERPs crash when 5,000 students check results simultaneously. This architecture utilizes Supabase PgBouncer connection pooling and edge-network distribution, ensuring 100% uptime under extreme load.

- **Edge-Middleware RBAC:** Role-Based Access Control is enforced at the Edge runtime using encrypted `jose` JSON Web Tokens. Unauthorized access attempts are violently rejected before the server even processes the request.
- **Multi-Factor OTP:** Faculty and Administrator accounts are protected by mandatory, mathematically expiring 6-digit SMS OTP gateways.

- **Strict XSS & CSRF Prevention:** Implemented strict Content-Security-Policies (CSP), cryptographic nonces, and X-Frame-Options: DENY to neutralize ambient credential hijacking and payload tampering.
- **Transactional Integrity:** Crucial database mutations (like attendance and grading) utilize strict PostgreSQL SERIALIZABLE isolation levels to mathematically eliminate read-modify-write data race conditions.

2. Executive Commercial Proposal

To provide KSRM with predictable, amortized capital expenditure, this digital transformation is structured across a 5-year managed infrastructure partnership, ensuring a lower Total Cost of Ownership (TCO) compared to maintaining legacy on-premise hardware.

2.1 5-Year Financial Structure (TCO)

PHASE	DELIVERABLES & DESCRIPTION	INVESTMENT VALUE (INR)
Year 1 (CapEx) Initial Deployment	Core IP License, Deployment & Migration Includes the full UMS software suite (all modules), legacy database extraction/cleansing, cloud provisioning, and campus-wide staff onboarding.	₹24,00,000 (One-Time Setup)
Years 2-5 (OpEx) Managed SLA	Premium Managed Cloud & Security SLA Includes AWS/Vercel enterprise hosting, Supabase connection pooling, automated database backups, DPDP privacy updates, and 24/7 exam-day support.	₹6,50,000 / Year (₹26,00,000 over 4 yrs)
5-Year Total Contract Value (TCV)		₹50,00,000

** All pricing is exclusive of applicable GST and third-party payment gateway transaction fees.*

3. Master Services Agreement (MSA)

This Master Services Agreement ("Agreement") outlines the commercial terms between the Provider ("Agency") and K.S.R.M. College of Engineering ("Client").

3.1 Term of Agreement

This Agreement shall remain in full force for an initial fixed term of **five (5) academic years** from the Effective Date. Upon expiration, this Agreement may be renewed upon mutual written consent.

3.2 License & Cloud Ownership

The UMS is provided as a fully managed Software-as-a-Service (SaaS). The application will be hosted exclusively on the Provider's highly-secure cloud architecture to prevent local tampering. **The Client retains absolute ownership of all generated institutional and student data.**

3.3 Payment Schedule (Year 1 CapEx)

The Year 1 deployment fee of ₹24,00,000 shall be invoiced according to the following performance milestones:

- **30% (₹7,20,000):** Upon contract execution and commencement of legacy data migration.
- **40% (₹9,60,000):** Upon successful User Acceptance Testing (UAT) of the EMS and Finance modules.
- **30% (₹7,20,000):** Upon official campus-wide "Go-Live" and completion of staff training.

3.4 Annual Managed SLA (OpEx)

Commencing on the first anniversary of the "Go-Live" date, the Client shall be invoiced the Annual Managed SLA fee of ₹6,50,000. Payment is mandatory to maintain cloud hosting, security patching, and technical support. Failure to remit within 30 days may result in suspension of services.

For K.S.R.M. College of Engineering:

Name / Title:

Date:

For the Provider:

Name: Andluru Sai Punith Reddy

Title: Lead Enterprise Architect

Date:

4. Enterprise Service Level Agreement (SLA)

This document establishes the guaranteed service levels, disaster recovery protocols, and emergency response times provided to KSRM under the active SLA.

4.1 Data Security & Disaster Recovery (DR)

- **Cryptography:** All institutional and financial data is encrypted at rest (AES-256) and in transit (TLS 1.3).
- **Automated Backups:** Redundant database snapshots are executed every six (6) hours and retained for 30 days. End-of-semester snapshots are archived permanently.
- **Recovery Time Objective (RTO):** In the event of catastrophic failure, the system is guaranteed to be restored from the latest snapshot within 4 hours.

4.2 Incident Response Matrix

Support tickets raised by KSRM IT Administrators will be resolved under strict timeframes during business hours (9:00 AM - 6:00 PM IST):

PRIORITY LEVEL	DESCRIPTION	GUARANTEED RESPONSE TIME
Severity 1 (Critical)	System outage or critical failure on Exam/ Results Day.	< 2 Hours (Hotfix Pipeline)
Severity 2 (High)	Major module failure (e.g., Razorpay/Tally sync issue).	< 4 Hours
Severity 3 (Standard)	UI bugs, reporting tweaks, or minor portal issues.	< 24 Hours

Compliance Notice: The architecture is designed to strictly align with the Digital Personal Data Protection Act, 2023 (DPDP). The Provider maintains security protocols, while the Client assumes responsibility for internal authorization and data governance by its staff.

Acknowledged by KSRM IT Dept:

Name / Title:

Date: